



CS 111 - Lab

Programming Language 2
Computer Science Department
2026

Lab 9 Exception handling and Files

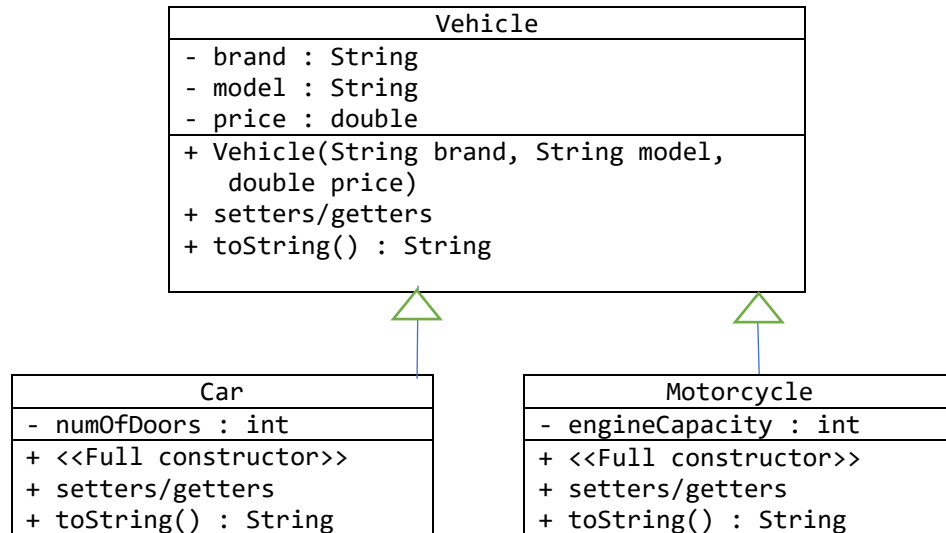
Lab Objectives:

In this lab, a student will practice

- | | |
|---|---|
| <ul style="list-style-type: none">• Reading part of a UML class diagram• Creating a class.• Declaring instance variables, constructor and accessors and mutators methods.• Implementing is-a relationship. | <ul style="list-style-type: none">• Using throw, throws and catch• Reading from a file• Writing to a file• Writing a test application to demonstrate the capabilities of another classes. |
|---|---|

Lab Exercise:

Given the following UML, implement all the classes in the same package with all needed instance variables, accessor and mutator methods, and constructors. Then write a test application to demonstrate the classes' capabilities.



Important Instructions

- Use NetBeans to create a new Java project named: CS111_YourName_YourID
- Make sure to save your work regularly. You are responsible for keeping a backup copy in case of any technical issues. No extra time will be given.
- Upload the zipped folder of your entire project.
- Copy and paste your code into a Word document, then save it.
- Upload both files (the zipped project folder and the Word document) in the same Blackboard submission link.

Part 1: UML Implementation

Given the UML (Vehicle, Car, Motorcycle):

Implement all classes exactly as shown in the UML diagram.

Requirements:

- Use inheritance (extends)
- Include all attributes, constructors, getters and setters
- Override `toString()` in subclasses

Part 2: Exception Handling

General Requirement

The program must be designed to handle exceptions properly.

You will implement the exception handling logic later in Part 3 inside the main method.

A. Use throw inside setter methods

- In Vehicle:
 - setPrice(double price)
 - If price < 0, throw IllegalArgumentException
- In Car:
 - setNumOfDoors(int numOfDoors)
 - If numOfDoors ≤ 0, throw IllegalArgumentException
- In Motorcycle:
 - setEngineCapacity(int engineCapacity)
 - If engineCapacity ≤ 0, throw IllegalArgumentException

B. Use throws in constructors

Constructors must declare possible exceptions (e.g., IllegalArgumentException).

Part 3: File Handling and Exception Handling in main

Input File: vehicles.txt

Students must prepare the input file in one of the following ways:

- Place the file inside the NetBeans project folder (recommended), or
- Create the file in any location and use the full file path in your code

Example:

```
new File("vehicles.txt")
```

or

```
new File("C:\\Users\\YourName\\Desktop\\vehicles.txt")
```

File Content:

C Toyota Camry 25000 4

M Yamaha R1 18000 1000

C Honda Civic 22000 4

M Kawasaki Ninja 15000 650

Tasks

1.Read Data

- Use Scanner to read data from the file
- For each line, read the first character (type)

Then:

- If the type is C:
 - Create a Car object using the data in the line
 - Add the object to the ArrayList<Vehicle>
- If the type is M:
 - Create a Motorcycle object using the data in the line
 - Add the object to the ArrayList<Vehicle>

- If the type is not C or M:
 - Throw IllegalArgumentException

2. Write to File

Write the data in the ArrayList to output.txt in reverse order without modifying the list.

3. Exception Handling in main

You must enclose the file processing code inside a try block and handle exceptions using multiple catch blocks.

Structure:

```
try {
    // file reading and writing code
}
catch (...) {
    // handle exception
}
```

You must handle exceptions in the following order (Specific → General):

1. When the file does not exist or cannot be opened.
2. When the data type in the file is incorrect.
3. For invalid values (generated from setter methods using throw).
4. For general input/output errors during file operations.
5. A general catch block for any unexpected errors.

Output Errors

All error messages must be printed using:

`System.err.println(...)`

Expected Output

M Kawasaki Ninja 15000 650

C Honda Civic 22000 4

M Yamaha R1 18000 1000

C Toyota Camry 25000 4